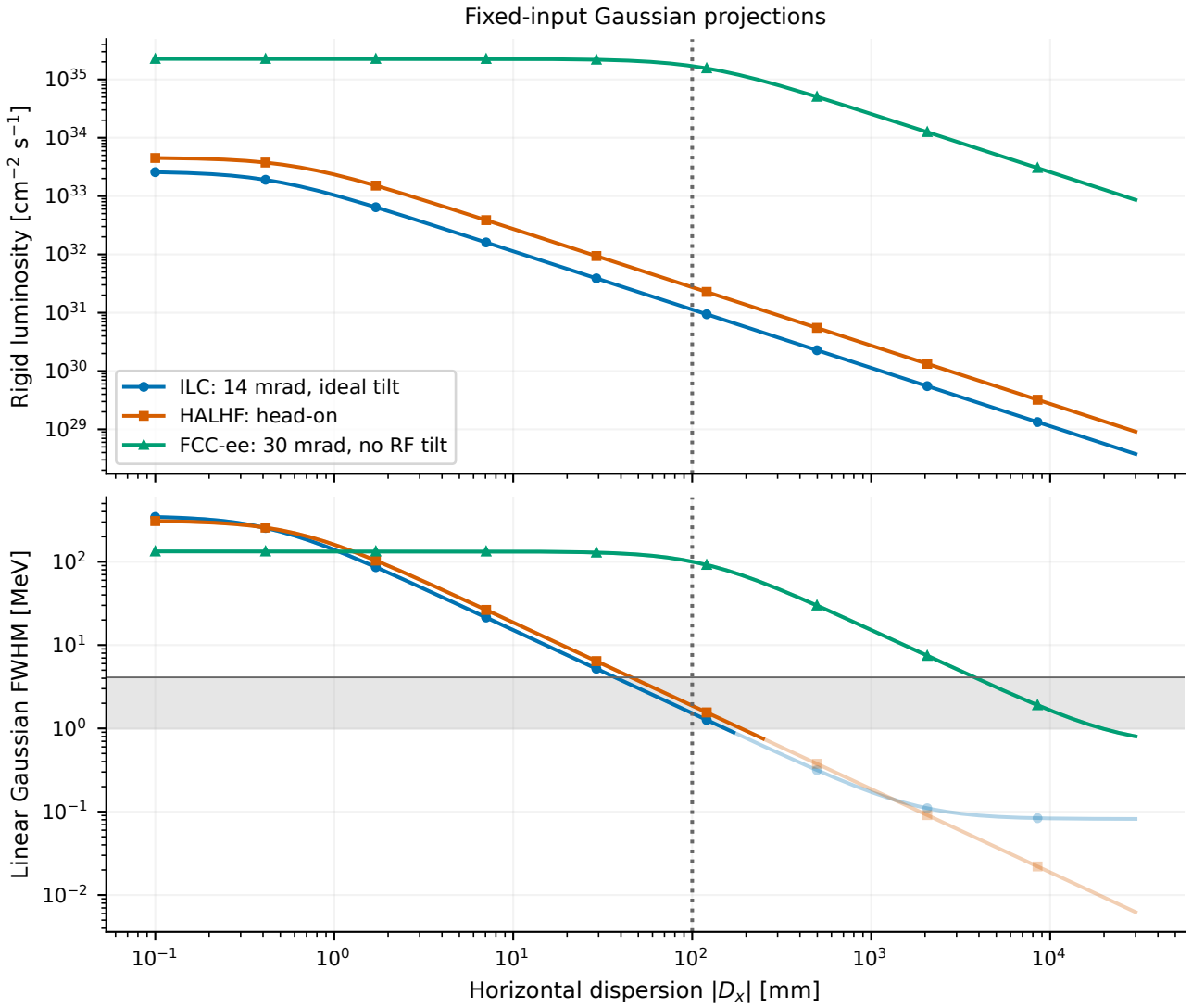

Omitted: radiation, disruption, RF errors, transported optics and waist/hourglass evolution.
FCC-ee L(100 mm)/L(0): rigid model 0.753; native 0.483. Matched controls are required.
Faint width segments: nonlinear RMS check differs by >5%; beyond 100 mm is projection only.